HOUSTON CITY COLLEGE

AI & Robotics Program

ITAI 4374 — Neuroscience as a Model for AI

MIDTERM & FINAL PROJECT INSTRUCTIONS

Bio-Inspired AI Audit

How well does real-world AI actually learn from the brain?

Semester	Spring 2026
Instructor	Patricia McManus
Midterm Due	March 25
Final Due	May 6
Midterm Weight	20% of Final Grade
Final Weight	30% of Final Grade

The Project

All semester you have been learning how the brain works — how it sees, remembers, pays attention, learns, and makes decisions — and how those same ideas show up (or fail to show up) in AI systems. Now you put that knowledge to work.

You will pick a **real AI product** that you can actually use, then run a series of hands-on experiments designed to answer one question: **How well does this AI system actually reflect what we know about the brain?**

Think of yourself as a quality inspector on a factory floor. Your job is to test the product, document what works and what does not, and write up professional recommendations for improvement.

How It Works

The midterm is not a separate assignment — it is the first half of your final project. Everything you build for the midterm carries forward and expands into the final.

Midterm — You pick your AI system, run your first round of experiments covering at least three course modules, and deliver a working audit-in-progress with documented evidence.

Final — You expand your midterm to cover at least five modules, add a Brain vs. AI Scorecard and a brain-inspired redesign section, and record a short video demonstrating your key experiments.

Choose Your AI System

Pick any AI product you can **directly interact with** and run experiments on. It must be something you can access repeatedly throughout the semester. Free or freemium access is fine. If you want to audit something not listed below, check with me (instructor) for approval.

Category	Examples
Chatbots & Assistants	ChatGPT, Claude, Gemini, Copilot, Perplexity, Meta AI, Grok
Image Generators	DALL·E, Midjourney, Stable Diffusion, Adobe Firefly, Ideogram
Voice Assistants	Siri, Alexa, Google Assistant
Code Assistants	GitHub Copilot, Cursor, Replit AI, Codex
Recommendation Engines	Spotify, YouTube, Netflix, TikTok
Vision & Detection	Google Lens, plant ID apps, Hugging Face vision models
Music / Audio AI	Suno, Udio, ElevenLabs, NotebookLM Audio
Specialized Tools	Grammarly, AI tutoring tools, medical AI demos

Report Format Requirements

Both the midterm and final reports must be written as **professional documents in prose format**. This means full paragraphs with clear topic sentences, supporting details, and transitions. Bullets point or numbers list may be use occasionally, but do write the whole doc in bullet points, numbered lists or outlines.

Write as if you are submitting an evaluation report to a manager or client. Each section should flow naturally from one idea to the next. Use headings to organize your sections, but the content under each heading should be written in complete, well-structured paragraphs.

When describing your experiments, write them as a narrative: explain what you set out to test, describe what you did step by step, present what you observed, and then discuss what it means in terms of the neuroscience from the course. Include screenshots and recordings as supporting evidence embedded within or appended to your report.

> **Why Prose Matters**
>
> In the workplace, you will write reports, evaluations, and recommendations. Professionals do not submit bullet-point lists to clients. They write clearly, organize their thinking, and present evidence in a readable format. This project is practice for that.

Midterm: Pilot Audit

The midterm is your first working pass at auditing the AI system through a neuroscience lens. You are delivering real findings, not a plan or proposal.

What You Deliver

Audit Report (4–6 pages, prose format)

Your midterm report should include the following sections, all written in professional prose:

System Profile: Describe the AI system you chose, what it does, who it is for, how you are accessing it, and why you selected it. This section should give the reader a clear understanding of the product before you begin discussing your experiments.

Experiment Results: Present at least **three (3) hands-on experiments**, each connected to a different course module. For each experiment, describe the brain-inspired concept you were testing, explain what you did step by step, present your results with supporting evidence (screenshots, chat logs, recordings), and discuss how your findings connect to the neuroscience from the course. Did the AI behave like the brain or not? How and why?

Next Steps: A brief paragraph outlining which additional modules you plan to cover in the final and any adjustments based on what you learned in the pilot.

Midterm Naming Convention

Group submissions: MD_GroupName_SubmitterName_ITAI4374

Example: MD_Group1_Joe_Smith_ITAI4374

Individual submissions: MD_Solo_YourName_ITAI4374

Example: MD_Solo_Mary_Doe_ITAI4374

Experiment Ideas by Module

These are starting points to spark your thinking. The best audits test things nobody else thought of.

Module	Experiment Ideas
Module 02 Brain Architecture	Test parallel vs. sequential processing. Can it multitask like the brain? Try giving it multiple instructions simultaneously and see how it handles them.
Module 03 Perception & Vision	Test visual edge cases: optical illusions, ambiguous images, rotated or partially hidden objects. Does it see like a human or fall apart?
Module 04 Memory & Learning	Test short-term vs. long-term memory. Does it hold context? What happens when you return the next day? Can you create false memories in it?
Module 05 Motor Systems	For coding or robotics AI: test action sequences, error correction, and planning. Does it plan ahead or just react step by step?
Module 06 Emotion & Motivation	Test emotional understanding. Can it detect sarcasm? Does it adjust to your tone? Try to manipulate its behavior through emotional framing.
Module 07 Attention & Consciousness	Test selective attention: can it focus on what matters and ignore noise? Overload it with information. Test for anything resembling self-awareness.

Frequently Asked Questions — Midterm

For the midterm, do I need to cover modules we have not reached yet in class? No. The midterm only requires experiments from modules covered so far in the semester. You are not expected to test concepts we have not yet discussed.

Can two students audit the same AI system? Yes, but your experiments and findings must be completely independent. Two students can both audit ChatGPT, for example, but each student should design their own experiments, collect their own evidence, and write their own analysis. If two reports look similar, both will be reviewed for academic integrity.

What if my experiment does not produce interesting results? That is a perfectly valid finding. If the AI system handles something well and matches what the brain does, say so and explain why. If the AI just gives a boring, predictable response, that tells you something too — perhaps it lacks the variability or creativity that the brain's mechanisms produce. The quality of your analysis matters more than whether you found something dramatic.

What format should my report be in? Submit as a Word document (.docx) or PDF. Write in professional prose — full paragraphs, not bullet points or outlines. Use headings to organize your sections, but the content under each heading should be written in complete, well-structured paragraphs. Think of it as a report you would submit to a manager, not a list of notes.

What counts as evidence? Screenshots, screen recordings, exported chat logs, saved AI outputs, before-and-after comparisons — anything that proves you actually ran the experiment and did not just describe it from memory. Label your evidence clearly so the instructor can match it to the corresponding section of your report.

Midterm Rubric (100 Points)

Category	Points	What Earns Full Credit
System Profile	15 pts	Clear, well-written description of the AI system and rationale for choosing it.
Experiment Design	25 pts	At least 3 experiments tied to 3 different modules. Each has a clear neuroscience hypothesis.
Evidence & Documentation	25 pts	Screenshots, recordings, or logs that prove experiments were actually conducted.
Neuroscience Connection	20 pts	Findings are meaningfully connected to specific course content with real understanding.
Next Steps	10 pts	Clear plan for expanding the audit in the final.
Professional Writing	5 pts	Prose format, well-organized, clearly written, reads like a professional report.

Final Project: Complete Audit + Video Demo

The final builds directly on your midterm. Incorporate instructor feedback, expand your experiments to cover more modules, and add your professional analysis and recommendations.

What You Deliver

Complete Audit Report (8–10 pages, prose format)

Your final report expands the midterm into a comprehensive evaluation. All sections must be written in professional prose:

Executive Summary — A one-page overview of the system you audited, your approach, your key findings, and your overall verdict. A reader should understand the entire audit from this section alone.

System Profile — Updated from midterm with any new insights from deeper use.

Experiment Results — A minimum of **five (5) course modules** covered with distinct experiments. Each experiment is written as a narrative: what you tested, what you did, what you found, and what it means. Incorporate your midterm experiments (revised based on feedback) alongside new ones.

Brain vs. AI Scorecard — A summary table rating how well the AI performs on each brain-inspired dimension you tested. Use a simple scale (Strong, Partial, Weak, or Missing) with a one-sentence justification for each. Follow the table with a paragraph discussing overall patterns and surprises.

Brain-Inspired Redesign — Your professional recommendations. If you could redesign this AI system using neuroscience principles from the course, what would you change? Be specific — connect each recommendation to a brain mechanism and explain how it would improve the system.

Video Demo (5–7 minutes)

Record a **screen-capture video with voiceover** demonstrating your most interesting experiments. This is not a narrated slideshow — show the AI system on screen, run your experiments, and walk the viewer through what happened and why it matters. Upload your video to the course LMS by the final due date.

You can record using any free tool: OBS Studio, Loom, Zoom (start a meeting alone and record), PowerPoint screen recording, QuickTime on Mac, or Xbox Game Bar on Windows.

Final Naming Convention

Group submissions: FN_GroupName_SubmitterName_ITAI4374

Example: FN_Group1_Joe_Smith_ITAI4374

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Example: FN_Solo_Mary_Doe_ITAI4374

Brain vs. AI Scorecard Template

Use this format or similar in your final report. (This is an example only you have all 14 modules to choose

Dimension	Rating	Module	Evidence Summary
Selective Attention	Strong / Partial / Weak / Missing	Module 07	One-sentence justification
Memory Consolidation	Strong / Partial / Weak / Missing	Module 04	One-sentence justification
Emotional Processing	Strong / Partial / Weak / Missing	Module 06	One-sentence justification
Error Correction	Strong / Partial / Weak / Missing	Module 05	One-sentence justification
Visual Perception	Strong / Partial / Weak / Missing	Module 03	One-sentence justification

Frequently Asked Questions — Final

Do I need one scorecard dimension for every single module? No. Your scorecard dimensions should reflect the specific capabilities you actually tested, not a checklist of modules. You may have multiple dimensions from the same module (for example, Module 04 could produce both a "Short-Term Context" row and a "Long-Term Memory" row), or a single experiment that spans multiple modules. The minimum requirement is five modules covered — but the scorecard rows depend on what you tested.

Can I change my AI system after the midterm? Only with instructor approval, and only if there is a strong reason (for example, the tool shut down or moved behind a paywall). Switching systems means starting your experiments over, so think carefully before requesting a change. If you are unsure, schedule an appointment with the instructor.

How long should my video be? Five to seven minutes. Focus on your two or three most interesting experiments. You do not need to cover everything from your report, pick the findings that are most surprising, most revealing, or most fun to demonstrate on screen.

Do I need to redo my midterm experiments for the final? Not from scratch. Incorporate your midterm experiments into the final report, revised based on instructor feedback. Then add new experiments to reach the five-module minimum. The final builds on the midterm it does not replace it.

What if the AI system updated between my midterm and final? That is actually interesting material for your report. If the system changed, document what is different and discuss whether the update made it more or less brain-like. AI systems evolving mid-audit is a real-world scenario, not a problem.

Final Project Rubric (100 Points)

Category	Points	What Earns Full Credit
Executive Summary	5 pts	Clear, concise overview. A reader understands the whole audit from this page.
Experiment Coverage	20 pts	Minimum 5 modules with distinct, well-designed experiments.
Experiment Quality	20 pts	Creative, well-controlled experiments that genuinely test brain-inspired principles.
Evidence & Documentation	15 pts	Every claim is backed by documented proof: screenshots, recordings, or logs.
Neuroscience Depth	15 pts	Deep, accurate connections to course content. Real understanding, not surface comparisons.
Brain-Inspired Redesign	10 pts	Specific, actionable recommendations grounded in neuroscience.
Video Demo	10 pts	Engaging screen-capture demo showing experiments in action with clear narration.
Professional Writing	5 pts	Prose format, well-organized, reads like a professional evaluation report.

Academic Integrity

This project requires original, hands-on experimentation. Run your own experiments and document them honestly. If an experiment did not produce interesting results, say so — that is a valid finding. Cite your sources when referencing course material or external information. You may use AI tools to help with formatting or grammar, but the analysis, conclusions, and recommendations must be your own thinking. If your audit reads like it was written by the same AI you are auditing, that is a problem.

Tips for Success

Be a detective, not a tourist. Do not just use the AI normally and describe what happened. Design experiments with a specific neuroscience hypothesis in mind. "I wonder if this chatbot has anything resembling selective attention" is a hypothesis. "I used ChatGPT and it was pretty good" is not.

Try to break it. The most interesting findings come from pushing the system to its limits. Contradict it, overload it, feed it noise. The places where AI fails are often the most neuroscience-rich moments in your audit.

Compare brain and machine side by side. For each experiment, describe what the brain would do in the same situation, then show what the AI actually did. The gap between those two is where your analysis lives.

Document as you go. Take screenshots and recordings during every experiment, not after. You cannot recreate AI interactions, especially with systems that generate different responses each time.

If you need help with experiment design or your neuroscience connections, **schedule an appointment with the instructor**. Getting feedback early always leads to better work.

Good luck, have fun, and break some AI.

Patricia